

Physics Practical Skills Pre-Test Question Bank

Programme: Teacher Training Programme

Focus: Class 11 and 12 Physics Practical & Pedagogical Skills

Purpose: Diagnostic Pre-Test Assessment

Section 1: Multiple Choice Questions

Question 1: While using a Vernier calliper, a trainee teacher observes that the zero of the Vernier scale lies to the right of the zero of the main scale when the jaws are in contact. What type of error is this and how must it be handled?

- A. Positive zero error; it should be added to the final observed reading.
- B. Positive zero error; it should be subtracted from the final observed reading.
- C. Negative zero error; it should be subtracted from the final observed reading.
- D. Negative zero error; it should be added to the final observed reading.

Question 2: What is the primary cause of backlash error in a screw gauge or spherometer, and how can students be taught to minimize its impact during an experiment?

- A. Incorrect alignment of the main scale reference line; avoid it by recalculating the index error.
- B. Non-uniformity of the measured wire; avoid it by taking readings at multiple perpendicular positions along the wire.
- C. Wear and tear of the screw threads; avoid it by turning the thimble in one direction only while taking a particular reading.
- D. Thermal expansion of the metal frame; avoid it by minimizing direct hand contact with the gauge.

Question 3: In a spherometer experiment to determine the radius of curvature R of a spherical surface, the formula used is $R = (l^2 / 6h) + (h / 2)$. What does the variable l represent?

- A. The total diameter of the lens or curved surface under measurement.
- B. The vertical distance moved by the central screw from the flat plane.
- C. The pitch of the micrometer screw thread on the central pillar.
- D. The mean distance between any two fixed outer legs of the spherometer.

Question 4: The mass and side length of a uniform copper cube are measured with maximum percentage errors of 1.5% and 1.0% respectively. What is the maximum percentage error in the calculated density of the cube?

- A. 4.5%
- B. 2.5%

- C. 3.5%
- D. 5.5%

Question 5: Why is a small, high-density spherical brass bob preferred over a large wooden bob in a simple pendulum experiment to determine the acceleration due to gravity (g)?

- A. To increase the period of oscillation so it can be timed manually with higher precision.
- B. To minimize air resistance effects and maintain a consistent, compact center of gravity.
- C. To reduce the physical tension on the supporting string thread.
- D. To allow the pendulum to oscillate with large angular amplitudes safely.

Question 6: In a resonance tube experiment, the first resonance length is l_1 and the second resonance length is l_2 . What is the standard expression for the end correction e ?

- A. $e = l_2 - 2l_1$
- B. $e = (3l_1 - l_2) / 2$
- C. $e = (l_2 - 3l_1) / 2$
- D. $e = (l_2 - l_1) / 2$

Question 7: Why are students strongly advised to configure experiment variables to obtain the null balance point near the middle of a metre bridge wire (between 40 cm and 60 cm)?

- A. To maximize the total current flowing through the central galvanometer circuit.
- B. To prevent the wire from developing permanent kinks or localized melting due to hot spots.
- C. To completely eliminate the intrinsic resistance of the heavy copper end strips.
- D. To maximize the measurement sensitivity of the Wheatstone bridge layout and minimize percentage errors.

Question 8: A student sets up a potentiometer experiment and notices that the null deflection point is consistently reached very close to the zero end of the wire. Which procedural modification shifts the balance point toward the middle of the wire?

- A. Decreasing the current in the primary driver circuit by increasing the series rheostat resistance.
- B. Increasing the current in the primary driver circuit.
- C. Decreasing the overall length of the potentiometer wire layout.
- D. Connecting a high resistance box in series with the protective galvanometer.

Question 9: During a cell EMF comparison experiment using a potentiometer, the galvanometer shows deflection in only one direction at both ends of the wire. What is the most likely structural error?

- A. The positive terminal of the secondary circuit cell is joined to the final terminal of the wire.
- B. The EMF of the primary driver battery is less than the individual EMFs of the secondary cells being measured.
- C. The internal resistance of the secondary test cell is higher than the wire resistance.

D. The galvanometer sensitivity is adjusted too high for the scale.

Question 10: In a terminal velocity experiment using Stokes' Law to find the viscosity of glycerin, why must the glass tube container be significantly wide compared to the diameter of the dropping steel ball?

- A. To prevent the liquid from expanding or overflowing due to fluid displacement.
- B. To ensure that the dropping ball reaches its steady terminal velocity rapidly.
- C. To minimize boundary wall effects that introduce additional upward drag forces on the falling ball.
- D. To allow the sphere to fall in a curved path for clear side profile viewing.

Question 11: What is the consequence of oil or greasy contamination inside a capillary tube when attempting to measure the surface tension of water using the capillary rise method?

- A. It forces a concave meniscus to immediately flip into a highly stable convex mercury-like shape.
- B. It decreases the contact angle, forcing an increased height of capillary rise.
- C. It keeps the height constant but changes the density of the fluid meniscus layer.
- D. It increases the contact angle, resulting in a significantly reduced height of capillary rise.

Question 12: For a real image formed by a convex lens on an optical bench, which of the following graph coordinates produces a straight line that can be easily used to read the focal length from its intercepts?

- A. A plot of $1/v$ on the Y-axis versus $1/u$ on the X-axis (using Cartesian sign conventions).
- B. A plot of direct image distance v versus direct object distance u .
- C. A plot of u^2 versus v^2 .
- D. A plot of u on the Y-axis versus $1/v$ on the X-axis.

Question 13: How can a trainee verify that parallax has been completely removed between an object pin and the real image pin on an optical bench?

- A. By confirming that both pins are adjusted to the exact same vertical height from the base.
- B. By shifting the eye sideways; both pins must appear to move together without any relative displacement between their tips.
- C. By bringing the pins close enough together that their physical tips touch.
- D. By closing one eye and verifying that the image pin appears upright and sharp.

Question 14: When plotting the relationship between the angle of incidence (i) and the angle of deviation (δ) for a glass prism, what is the characteristic shape of the resulting curve?

- A. It remains entirely constant across all changes to the angle of incidence.
- B. It increases linearly as the angle of incidence increases.
- C. It decreases initially to a minimum value and then increases as the angle of incidence continues to rise.
- D. It forms a continuously decreasing curve that approaches zero asymptotically.

Question 15: To convert a galvanometer of coil resistance G into an ammeter of a desired range 0 to I , a shunt resistance S is connected in parallel. If the full-scale deflection current is I_g ,

what is the correct formula for S?

- A. $S = (I - I_g) \times G$
- B. $S = ((I - I_g) \times G) / I_g$
- C. $S = (I_g \times G) / I$
- D. $S = (I - I_g) \times G / (I - I_g)$

Question 16: When converting a galvanometer into a voltmeter, a high resistance R is connected in series. If the desired voltage range is V and the galvanometer resistance is G, what is the mathematically sound choice for R?

- A. $R = (V / I_g) - G$
- B. $R = (I_g / V) - G$
- C. $R = (V - G) / I_g$
- D. $R = V \times I_g - G$

Question 17: In a concave mirror experiment to find the focal length, what happens to the image if the lower half of the mirror's reflecting surface is covered with an opaque black paper?

- A. Only the upper half of the object's image is visible on the screen.
- B. A full, complete image continues to form, but its total brightness/intensity is reduced.
- C. Only the lower half of the object's image is visible on the screen.
- D. The image disappears entirely because the principal axis is blocked.

Question 18: When validating Newton's Law of Cooling, a graph of $\ln(\theta - \theta_0)$ versus time (t) is plotted (where θ is the container temperature and θ_0 is room temperature). What should the shape of this graph be?

- A. A perfect parabola that opens upward.
- B. A straight line passing through the origin with a positive slope.
- C. A straight line with a negative slope.
- D. An exponential curve that approaches the horizontal time axis asymptotically.

Question 19: In a sonometer experiment, if the wire tension T and linear mass density m are held constant, what relationship should be plotted to yield a straight line passing through the origin?

- A. The square of frequency (f^2) directly versus the length (l).
- B. Frequency (f) directly versus the vibrating length (l).
- C. Frequency (f) versus the square root of length.
- D. Frequency (f) on the Y-axis versus the reciprocal of length (1/l) on the X-axis.

Question 20: In the forward bias characteristics of a p-n junction diode, what is the region before the current begins to rise sharply called, and what electrical barrier must be overcome?

- A. Knee voltage region; the external forward voltage must overcome the internal potential barrier of the depletion layer.
- B. Breakdown voltage region; it corresponds to the thermal ionization of covalent bonds.

- C. Ohmic linear conduction region; it satisfies a fixed resistance ratio.
- D. Saturation current plateau; it is limited by minority carrier diffusion.

Question 21: Why is it standard laboratory practice to disconnect the plug key in an electrical circuit between taking consecutive sets of readings?

- A. Continuous current causes the battery's zero calibration to drift over time.
- B. Continuous current flow heats up the wires and resistors, which changes their resistance and introduces systematic errors.
- C. The internal coils of the ammeters and voltmeters may become permanently magnetized.
- D. To allow accumulated static charge on the insulation surfaces to dissipate.

Question 22: A student uses a travelling microscope to measure the refractive index of a glass slab. They record three vertical positions: the real bottom mark (R_1), the apparent bottom mark viewed through the slab (R_2), and a powder mark on the top surface (R_3). What is the correct expression for the refractive index μ ?

- A. $\mu = (R_2 - R_1) / (R_3 - R_1)$
- B. $\mu = (R_3 - R_2) / (R_3 - R_1)$
- C. $\mu = (R_3 - R_1) / (R_2 - R_1)$
- D. $\mu = (R_3 - R_1) / (R_2 - R_1)$

Question 23: A spherometer screw advances by 2 mm over 4 complete rotations. If the circular scale contains 100 equal divisions, what is the calculated least count of this instrument?

- A. 0.05 mm
- B. 0.01 mm
- C. 0.002 mm
- D. 0.005 mm

Question 24: When analyzing an experiment governed by the physical law $y = kx^2$, what variables should be plotted on the axes to determine the constant value k directly from the slope of a linear graph?

- A. Plot y on the Y-axis versus x^2 on the X-axis.
- B. Plot y directly versus x .
- C. Plot $\sqrt{y}$ versus x^2 .
- D. Plot y versus $1/x$.

Question 25: What are 'end resistances' or 'end errors' in a metre bridge experiment, and how can they be minimized during calculations?

- A. Resistances from the internal components of the driver cell; they are minimized by using a

battery with a higher voltage output.

- B. Resistances due to the terminating copper strips and wire soldering contacts; they are minimized by repeating measurements after interchanging the positions of the unknown and standard resistors.
- C. Inaccuracies from variations in the bridge wire's cross-section; they are minimized by sliding the jockey firmly along the wire.
- D. Parallax errors at the extreme ends of the scale; they are minimized by using a magnifying glass to read the marks.

Section 2: Master Answer Key & Rationales

Q.No.	Correct Option	Pedagogical & Practical Rationale
1	B	When the Vernier zero is to the right of the main scale zero, it registers an initial positive offset value. The tool overestimates values, meaning the error must be subtracted from observed measurements.
2	C	Backlash arises from mechanical gaps or looseness between thread connections. Reversing rotational direction fails to move the spindle instantly, so reading trends must proceed in a single rotational direction.
3	D	The value l is the measured distance between the outer

Q.No.	Correct Option	Pedagogical & Practical Rationale
		supporting non-movable legs of the equilateral triangular spherometer frame.
4	A	Density error propagation follows standard rules: mass error + $3 \times$ length error = $1.5\% + 3(1.0\%) = 4.5\%$. This demonstrates scaling of errors with power constraints.
5	B	High-density dense materials isolate variables by keeping the physical center of mass perfectly localized and decreasing air drag effects compared to low-density objects.
6	C	Derived from matching consecutive nodes where anti-nodes extend slightly outside the edge boundary: $l_1 + e = \lambda/4$ and $l_2 + e = 3\lambda/4$. Solving for e yields $(l_2 - 3l_1)/2$.
7	D	The balancing bridge structure experiences the lowest fractional percentage

Q.No.	Correct Option	Pedagogical & Practical Rationale
		error when the ratio arms are uniform and closely matching, meaning balancing near the center yields optimum sensitivity.
8	A	Reducing primary driver loop current via series rheostat drops the overall potential gradient (voltage per unit wire length), demanding a longer wire stretch to match secondary test cell EMFs.
9	B	If the potential gradient profile across the bridge wire lacks sufficient magnitude compared to the secondary circuit's EMF value, a balancing zero cross-over can never occur on the wire.
10	C	Stokes' Law criteria assume a completely boundless fluid canvas. Proximity to container solid walls compresses liquid streamline channels, raising artificial retarding viscous drag forces.

Q.No.	Correct Option	Pedagogical & Practical Rationale
11	D	Grease drastically lowers glass-water cohesive/adhesive attraction parameters, pushing up the fluid contact angle θ . Since vertical lift tracks $\cos(\theta)$, the resulting liquid rise height drops significantly.
12	A	The reciprocal conversion maps standard non-linear lens equations directly into intercept straight-line format ($y = mx + c$), letting instructors map focal values linearly.
13	B	Parallax describes shifts in perspective caused by a change in viewpoint. If elements share an identical spatial focal coordinate, they lock together without showing relative separation when viewed from side to side.
14	C	Prism layout mechanics dictate an initial reduction in deviation toward a localized vertex minimum (the angle of minimum deviation),

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		following which additional increases in incidence lift deviation metrics.
15	D	Parallel connection setups hold equivalent potential variations. The parallel shunt formula balances currents: $I_g \times G = (I - I_g) \times S$, which simplifies to $S = (I_g \times G) / (I - I_g)$.
16	A	Ohm's law governing a full voltmeter loop dictates total resistance ($R + G$) must equal full-scale voltage capacity divided by device current threshold: $R + G = V / I_g$, leading to $R = (V / I_g) - G$.
17	B	Every open point on a reflecting plane captures rays from the entire object. Blanking parts reduces overall light gathering power (lowering image brightness) but does not trim the image structure.
18	C	Integrating Newton's cooling differential equation creates

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		a logarithmic linear plot: $\ln(\theta - \theta_0) = -kt + C$. This represents a straight line with a downward negative slope proportional to cooling coefficient values.
19	D	Fundamental frequency equations indicate that f varies inversely with string length l . Plotting f against the reciprocal term $(1/l)$ linearizes data points through the coordinate origin.
20	A	The knee voltage corresponds to the specific point where external forward bias fields overpower internal barrier potential frameworks generated across the p-n junction depletion zone.
21	B	Sustained circuit current triggers Joule heating inside wires, which alters intrinsic resistance variables. Pulling keys breaks circuits to preserve temperature stability and limit systematic

Q.No.	Correct Option	Pedagogical & Practical Rationale
		bias.
22	C	Refractive index equals the ratio of real slab depth to apparent slab depth. Geometrically, real thickness maps as $(R_3 - R_1)$ and apparent thickness maps as $(R_3 - R_2)$, yielding $\mu = (R_3 - R_1) / (R_3 - R_2)$.
23	D	Pitch equals 2 mm divided by 4 rotations, which matches 0.5 mm per turn. The instrument least count equals pitch divided by total scale divisions: $0.5 \text{ mm} / 100 = 0.005 \text{ mm}$.
24	A	Mapping y against the grouped exponential factor (x^2) satisfies straight-line equations ($Y = mX$), enabling direct collection of the constant k as a linear gradient.
25	B	End errors relate to unexpected additional resistance contributions from metal strip interfaces and corner solder gaps.

Q.No.	Correct Option	Pedagogical & Practical Rationale
		Interchanging component locations balances out these localized offsets across the bridge wire.